Customer Shopping Behavior Analysis

Data Analysis Report — Python, PostgreSQL & Power BI

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18

Key Features:

  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

Missing Data: 37 values in the Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python (see data.ipynb):

- Data Loading: Imported the dataset using pandas (pd.read_csv).
- Initial Exploration: Used df.head(), df.info(), and df.describe(include='all') to check structure and summary statistics.
- Missing Data Handling: Checked for null values with df.isnull().sum() (37 missing in Review Rating); imputed the missing values in the Review Rating column using the median rating of each product category (grouped by Category with .transform()).
- Column Standardization: Converted all column names to lowercase snake_case (spaces replaced with underscores); renamed purchase_amount_(usd) to purchase_amount for clarity.
- Feature Engineering:
  - Created age_group column by splitting customer ages into four equal-sized quantile bins (pd.qcut), labeled Young Adult, Adult, Middle-Aged, and Senior.
  - Created purchase_frequency_days column by mapping each frequency_of_purchases category (Weekly, Fortnightly, Monthly, Quarterly, Bi-Weekly, Annually, Every 3 Months) to its equivalent number of days.
- Data Consistency Check: Compared the discount_applied and promo_code_used columns and found them identical, so promo_code_used was dropped as a redundant column.
- Database Integration: Connected the Python script to PostgreSQL using SQLAlchemy and psycopg2, and loaded the cleaned DataFrame into a customer_data table for downstream SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue by Gender — Compared total revenue generated by male vs. female customers.

Gender	Revenue (USD)
Female	75,191
Male	157,890

2. High-Spending Discount Users — Customers who used discounts but still spent above the average purchase amount.

Customer ID	Purchase Amount (USD)
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62

Total rows returned: 839 (sample of first 10 shown above).

3. Top 5 Products by Rating — Products with the highest average review ratings.

Item Purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. Shipping Type Comparison — Average purchase amounts between Standard and Express shipping.

Shipping Type	Average Purchase Amount (USD)
Standard	58.46
Express	60.48

5. Subscribers vs. Non-Subscribers — Average spend and total revenue across subscription status.

Subscription Status	Total Customers	Avg. Spend (USD)	Total Revenue (USD)
Yes	1,053	59.49	62,645.00
No	2,847	59.87	170,436.00

6. Discount-Dependent Products — Products with the highest percentage of discounted purchases.

Item Purchased	Discount Rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer Segmentation — Customers classified into New, Returning, and Loyal segments based on purchase history.

Customer Segment	Number of Customers
Loyal	3,116
New	83
Returning	701

8. Top 3 Products per Category — Most purchased products within each category.

Rank	Category	Item Purchased	Total Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions — Whether customers with more than 5 purchases are more likely to subscribe.

Subscription Status	Repeat Buyers
No	2,518
Yes	958

10. Revenue by Age Group — Total revenue contribution of each age group.

Age Group	Total Revenue (USD)
Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually, including customer counts, average purchase amount, average review rating, subscription split, revenue and sales by category, and revenue and sales by age group, with interactive filters for subscription status, gender, category, and shipping type.

6. Business Recommendations

- Boost Subscriptions — Promote exclusive benefits for subscribers.
- Customer Loyalty Programs — Reward repeat buyers to move them into the “Loyal” segment.
- Review Discount Policy — Balance sales boosts with margin control.
- Product Positioning — Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing — Focus efforts on high-revenue age groups and express-shipping users.